Week 7: Assignment

Introduction to Wireless Communication

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Summary of Task:

In this task, we expanded our reinforcement learning framework from pure power control to joint resource management by combining **Transmit Power** (7 levels) and **Adaptive Modulation** (BPSK, QPSK, 16-QAM) into a unified 21-choice action space. We updated the custom Gymnasium environment to dynamically calculate theoretical Bit Error Rates (BER) and throughput in real time, shifting the reward function to maximize data rate while heavily penalizing corrupted transmissions (BER $> 10^{-3}$) and excess energy usage.

Using a Deep Q-Network (DQN), the agent successfully transitioned from blind exploration—where high error rates yielded negative rewards—to an optimal policy that consistently achieves clean, maximum-throughput transmissions. When evaluated against standard heuristic baselines, the trained agent dynamically adapts its power and modulation strategy based on user distance, proving that deep reinforcement learning can effectively master non-linear wireless physical layer trade-offs without requiring static manual rulebooks.

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